

Here's the complete list of all 38 research papers from our search, organized by category for your hackathon prep. I've starred (note: prioritized with " note: prioritized with **") the most relevant ones for your project.

Category 1: Reviews & Surveys (Start Here)

#	Paper	Year	Venue	Why Read It
1	**[A Review on Classification of Cardiac Arrhythmia in ICU Patients](https://www.semanticscholar.org/paper/f6a7a7c91b77093403bbfc6bb026edbl851ebbe)**	2024	–	Reviews 25 studies (2021-2024); great for understanding the landscape
2	**[Exploring the Future of AI-Enhanced Wearable Electronics for Real-Time Arrhythmia Detection](https://assets.cureusjournals.com/artifacts/upload/review_article/pdf/10001/20251210-98385-x6jvjr.pdf)**	2025	–	Comprehensive review of wearable sensor tech + AI integration
3	**[From ECG Signals to Intelligent Diagnosis: A Survey on DL-Based Arrhythmia Classification and Deployment](https://doi.org/10.48175/ijarsct-31004)**	2026	–	Most recent survey covering preprocessing, architectures, deployment

Category 2: CNN-Based ECG Classification (Core Models)

#	Paper	Year	Venue	Key Result
4	**[An Automated ECG Beat Classification System Using CNNs](https://www.semanticscholar.org/paper/6b2bac0c0f598d423d4bc0e9e25028d5d1795463)**	2016	ICITCS	**204 citations**;
5	[Sensor-Enhanced DL for ECG Arrhythmia Detection: Multiscale CNN + Self-Attentive GRU]	2024	IEEE Sensors Journal	99.63% accuracy; CNN+GRU+attention
6	[A Deep Learning Approach for Arrhythmia Detection Based on IoT Wearable Technology]	2022	BIWA	IoT + Fog computing pipeline for athletes
7	[Diagnosis and Classification of Cardiac Arrhythmias Using CNNs]	2023	ICECET	99.24% accuracy, 4-class on MIT-BIH
8	[A Novel Arrhythmia Classification Method Based on CNN Interpretation of ECG Images]	2019	ICIT	2D wavelet transform → image → CNN approach; 95.3%

| 9 | [Multiclass Heartbeat Classification using ECG Signals and CNNs](https://biblio.ugent.be/publication/8755702/file/8755705.pdf) – Qureshi et al. | 2022 | ICoDT2 | Multi-class beyond binary; MIT-BIH < Citations > 8 < / Citations > |

| 10 | [CNN-Based DNN Model for ECG Records Classification](https://www.semanticscholar.org/paper/cc92caeace54149b3df005ba6fa192aa256d20a5) – Khudhur et al. | 2023 | ICFNDS | 16-layer 1D-CNN; 5-class MIT-BIH < Citations > 9 < / Citations > |

| 11 | [Cardiac Arrhythmia Classification in ECG Signals with CNNs](https://doi.org/10.5220/0011682800003411) – Souza & Dantas | 2023 | ICPRAM | 5-class CNN on noisy signals < Citations > 10 < / Citations > |

| 12 | [Cardiac Arrhythmia Diagnosis based on Features Extraction and CNN](https://www.semanticscholar.org/paper/4dab3069b95872e1d8ad62cb0416753bfe0ec5c1) – Mohonta & Ali | 2021 | ICEEICT | CWT scalogram + AlexNet approach < Citations > 11 < / Citations > |

| 13 | [Classification of ECG Arrhythmia Using a CNN](https://www.semanticscholar.org/paper/5bfba0e3eb5a72be497737b1400098256b7a49d4) – Reethunandh et al. | 2023 | ICSSES | CNN vs SVM comparison (98.7% vs 94%) < Citations > 12 < / Citations > |

| 14 | [Modified Cardiac Arrhythmia Classification Using CNN Model](https://ejournal.yasin-alsys.org/AJSTEJA/article/download/5905/4704) – Abdulhafiz et al. | 2025 | AJSTEJA | AlexNet feature extractor + backprop |

| 15 | [Employing CNN Model in Predicting Heart Diseases](https://www.semanticscholar.org/paper/edc046865481e221b97ba57c03304e9a967d8842) – Alshreif et al. | 2024 | ISTJ | 1D-CNN on MIT-BIH; 97.7% accuracy |

Category 3: Edge Computing & Wearable Deployment (Most Relevant for Hackathon)

#	Paper	Year	Venue	Key Result
16	**[A Smart Wearable for Real-Time Cardiac Disease Detection Using Edge Computing AI](https://www.semanticscholar.org/paper/edcc1f8e69ded892be96ee3e9b31e1ea73bbff5c)**	2024	BSN	90.6% on-device; 6ms latency; 128mW power < Citations > 13 < / Citations >
17	**[Resource-Constrained On-Chip AI Classifier for Beat-by-Beat Arrhythmia Detection](https://www.mdpi.com/2079-9292/14/13/2654/pdf)**	2025	Electronics	ANN/CNN/LSTM comparison on ARM Cortex; 99.5% QRS F1 < Citations > 14 < / Citations >
18	**[Cardio-Edge: LSTM-Based ECG Classification on Wearable Devices](http://thesai.org/Downloads/Volumel6No7/Paper_86-Cardio_Edge_Hardware_Software_Co_design_Implementation.pdf)**	2025	IJACSA	FPGA co-design; 10x speedup; 99% accuracy < Citations > 15 < / Citations >
19	**[Automated Arrhythmia Classification: Lightweight Model on Ultra-Edge Device](https://doi.org/10.1109/access.2024.3473323)**	2024	IEEE Access	Knowledge distillation + pruning + quantization < Citations > 16 < / Citations >
20	[Real-Time Arrhythmia Detection on Wearables Using Quantized Transformers](https://mrcis.org/index.php/journal/article/download/174/290)	2025	MRCIS	8-bit quantized Transformer; noise-robust training < Citations > 17 < / Citations >

| 21 | [Real Time Arrhythmia Monitoring Based on Edge Computing and DNN] (<https://downloads.hindawi.com/journals/wcmc/2021/5563338.pdf>) – Liu et al. | 2021 | WCMC | 99.03% accuracy; CNN + attention ResNet at edge < citations > 18 < / citations > |

| 22 | [Edge-Optimized Weightless Neural Network for Low-Power Wearable Arrhythmia Detection] (<https://www.semanticscholar.org/paper/9fc9d79332523594cf99b26e636bc05b4e1d54e1>) – Pillai et al. | 2025 | DCAS | WNN on 45nm CMOS; ultra-low-power patch < citations > 19 < / citations > |

| 23 | [Lightweight Element-wise Product Enhanced NN for Arrhythmia Detection on Embedded Devices] (<https://www.semanticscholar.org/paper/d92566f66f6858427c83128f0fabe391f5560127>) – Tang et al. | 2025 | Comp. Methods Programs Biomed. | Depthwise separable CNN; tested on MIT-BIH, SVDB, INCART, PTB < citations > 20 < / citations > |

| 24 | [A Wearable ECG Monitoring System for Real-Time Arrhythmia Detection] (<https://www.semanticscholar.org/paper/d133aa0a95e339d64aebf20fdf6da495ebbcfe74>) – Clark et al. | 2018 | MWSCAS | Pan-Tompkins + wavelet on Raspberry Pi; email alerts < citations > 21 < / citations > |

Category 4: Transformer-Based Models (Cutting Edge)

#	Paper	Year	Venue	Key Result
25	**[Patient-Adaptive Beat-Wise Temporal Transformer for AF Classification] (https://www.semanticscholar.org/paper/29cc97f771f1cd37d4205fd1e2dd4343c40c2dfe)**	2024	IEEE Access	F1=0.982; patient-specific symbolic tokens < citations > 22 < / citations >
26	**[Automated Diagnosis of AF Using ECG Component-Aware Transformer] (https://www.semanticscholar.org/paper/a2a724a9ec12176cdfd9504celdd5188081c47f9)**	2022	Comp. Biol. Medicine	**24 citations**; segments P-wave/QRS/T-wave like a cardiologist < citations > 23 < / citations >
27	[Transformer-based Temporal Sequence Learners for Arrhythmia Classification] (https://www.semanticscholar.org/paper/33cde44ade4b87811c983e81ae002e455124263c)	2023	MBEC	DistilBERT on ECG; 99.92% accuracy on MIT-BIH < citations > 24 < / citations >
28	[TransMixer-AF: Real-Time AF Detection Using Single-Lead ECG] (https://doi.org/10.1109/access.2024.3467181)	2024	IEEE Access	ConvMixer + Transformer; GradCAM++ explainability < citations > 25 < / citations >
29	[Transformer-based ECG Classification for Early Detection of Cardiac Arrhythmias] (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1600855/pdf)	2025	Frontiers in Medicine	PCA + t-SNE + Transformer pipeline < citations > 26 < / citations >
30	[Deciphering Heartbeat Signatures: Vision Transformer for Explainable AF Detection] (https://arxiv.org/pdf/2402.09474)	2024	IEEE EMBC	Vision Transformer on Chapman-Shaoxing; explainability focus < citations > 27 < / citations >
31	[AF Classification and Prediction Explanation using Transformer NN] (https://www.semanticscholar.org/paper/b0767e08d03c7e066d23fe444710a1c1888f86f6)	2022	IEEE IJCNN	TNN highlights absent P-wave; F1=0.87 < citations > 28 < / citations >
32	[Self-supervised Learning for AF Detection with CNN-Transformer] (https://www.semanticscholar.org/paper/82e4b2e914275a72302eda)			

6a62c38e546a20dfb7) – Zou et al. | 2023 | IEEE BIBM | Self-supervised 'nextclip' pretraining; state-of-the-art results
< Citations > 29 < / Citations > |
| 33 | [EndAFD: End-to-End System for ECG Based AF Detection] (<https://www.semanticscholar.org/paper/1a53f6eab58772e910904efd438bflf6c92eaafa>) – Basak et al. | 2022 | SSRN | End-to-end Transformer pipeline; single-lead focus |

Category 5: Federated Learning & Privacy (Differentiator)

#	Paper	Year	Venue	Key Result
34	**[Asynchronous Federated Learning-based ECG Analysis for Arrhythmia Detection] (https://www.semanticscholar.org/paper/8404f9ddf18e474769e26d2d61d9f73585b84d4b)]**	2021	MeditCom	**50 citations**; async FL on ultra-edge nodes < Citations > 30 < / Citations >
35	**[Federated Learning for Arrhythmia Detection of Non-IID ECG] (https://www.semanticscholar.org/paper/b3cd99e528763381d9d47b0f098f75923ed90f8c)]**	2020	IEEE ICC	**40 citations**; addresses non-IID data challenge < Citations > 31 < / Citations >
36	[Privacy-preserving Federated Learning for In-Home Monitoring of Elderly] (https://www.semanticscholar.org/paper/05f445edae3b752aedbe1256cd1e66b567ba5dbc)	2023	AIxAS	PPFL applied to elderly ECG wearables < Citations > 32 < / Citations >
37	[Fusion of Personalized FL with Differential Privacy for Arrhythmia Diagnosis] (https://www.semanticscholar.org/paper/e5e3b71dcc17b76f8e3d0c3a09a0d226a9daed3d)	2024	-	GRU + differential privacy + federated learning

Category 6: Novel Sensor Hardware + AI

#	Paper	Year	Venue	Key Result
38	[AI-Enhanced Diagnosis of Atrial Arrhythmia Using 3D-Printed Origami ECG Sensors] (https://www.semanticscholar.org/paper/3f45e4209334a27111186492090038214f40c085)	2025	Biosensors & Bioelectronics	Reusable 3D-printed sensor; CWT+CNN for 11 rhythm types < Citations > 33 < / Citations >
39	[A Novel Multi-Modal DL Framework Using ECG and PPG Signals] (https://www.semanticscholar.org/paper/faac2d5a25e74ee24096a082562d94a21444202a)	2025	CISES	Fuses ECG + PPG with Transformer; optimized for wearables < Citations > 34 < / Citations >
40	[Real-time Cardiac Arrhythmia Detection Using ML and Wearable Devices] (https://healthinformaticsjournal.com/index.php/IJMI/article/download/36/37)	2024	Frontiers in Health Informatics	CNN + SVM hybrid; cloud-based pipeline
41	[Real Time Arrhythmia Detecting Wearable using a Novel DL Model] (https://www.semanticscholar.org/paper/9f9f3e98e340352292cf01e0523d5a63cdb34a05)	2020	ICPS	1D-CNN + LSTM; mobile notification system < Citations > 35 < / Citations >

Recommended Reading Order for Your Hackathon

1. Start with **Paper #3** (2026 survey) for a landscape overview
2. Then **Paper #4** (Zubair, 204 citations) for the foundational CNN approach you'll implement
3. Then **Papers #16-17** (Rahman & Morshed) for edge deployment architecture
4. Then **Paper #20** (Quantized Transformers) if you want a differentiating model choice
5. Then **Paper #34** (Sakib, 50 citations) if you want to add a federated learning angle
6. Finally **Paper #28** (TransMixer-AF) for the GradCAM++ explainability technique